MIDTERM ASSIGNMENT

Fresh Retail Decision Challenge

Data-Driven Decision Making

Item	Requirement
Assessment	Midterm Assignment
Mode	Pairs
Dataset	FreshRetailNet-50K
Dashboard	Power BI OR Tableau OR Python
Recommended duration	Approximately 2 weeks
Deadline	23:59, Sunday, 11 October 2026
Total marks	100

Core principle: Start with the business decision, use relevant and trustworthy evidence, then move from Observation → Insight → Decision.

1. Business Context

You are working as a Business Data Analyst for a fresh-food retail company operating across multiple cities and stores. Management is concerned about product availability, promotion effectiveness, store allocation, and demand changes caused by holidays, activities and weather conditions.

Some products sell strongly but frequently experience stockouts.

Discounts may stimulate demand, but management is unsure whether they are used effectively.

Observed sales may underestimate true demand when products are unavailable for part of the day.

Demand may differ across stores, cities, holidays, activities and weather conditions.

Main business question: What ONE action should management prioritise next month to improve retail performance?

2. Learning Outcomes

Frame a business problem before choosing analytical tools.

Understand and assess the quality and relevance of a real-world dataset.

Create meaningful KPIs and derived variables for a business question.

Analyse sales, stockouts, promotions and external factors.

Build an interactive dashboard that supports managerial decision making.

Distinguish Observation, Insight and Decision.

Make evidence-based recommendations and acknowledge data limitations.

3. Dataset: FreshRetailNet-50K

Use the FreshRetailNet-50K dataset provided/approved for the course. The dataset contains store-product-day observations with sales, inventory availability, promotion/activity indicators and weather-related variables.

Dataset Access

Official dataset: https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K

Students may use the train.parquet file for this assignment. If computing resources are limited, a reproducible subset may be used, but the filtering or sampling rule must be clearly documented in the Data Preparation Notes.

Do not silently alter or overwrite the original source data. Any filtering, sampling, cleaning, or transformation that affects the analysis must be documented.

Variable group	Typical examples	Business use
Identification	City, Store, Product	Segment and compare performance
Sales / Performance	Sale amount	Measure observed sales performance
Inventory	Stock status, stockout hours	Identify availability constraints
Promotion	Discount, activity flag	Evaluate promotional effectiveness
Calendar	Date, holiday	Identify temporal patterns
Weather	Temperature, rainfall, humidity, wind	Investigate external demand factors

Important: A zero or low observed sale does NOT automatically mean zero or low customer demand. Stockouts can censor observed sales.

4. Permitted Tools

For data preparation and analysis, students may use SQL, KNIME, Excel or Python. For the interactive dashboard, choose ONE of the following:

- Power BI
- Tableau
- Python dashboard (for example Streamlit, Plotly Dash, or an equivalent interactive solution)

The marking does not favour one dashboard platform over another. Marks are awarded for analytical reasoning, evidence, usability and decision support—not for the brand of tool used.

5. Assessment Structure

The assignment contains a COMMON CORE for all students, a DECISION TRACK chosen by each student, one UNIQUE ANALYTICAL QUESTION, and an optional DISTINCTION CHALLENGE.

Part A — Data Understanding and Business Relevance (10 marks)

A1. Explain what one row in the dataset represents.

A2. Classify important variables into Identification, Sales/Performance, Inventory, Promotion, Calendar and Weather.

A3. Identify which variables are most relevant to the management decision and explain why.

A4. Explain at least one situation in which Observed Sales ≠ True Demand.

A5. State one variable or data feature that could be misleading if interpreted without business context.

Part B — Data Quality and Preparation (10 marks)

Check missing values, duplicates, unusual values, date issues, discount values, weather availability and stock-status consistency.

Do not automatically delete unusual records. Explain the business meaning before deciding to keep, fix, flag, transform or remove them.

Create at least THREE useful derived variables, for example: HasStockout, StockoutRate, DiscountPercent, DayOfWeek, WeekendFlag, or SalesPerAvailableHour.

Document assumptions that may affect later interpretation.

Data issue	Action	Business/analytical reason
Example: unusual value		

Part C — Core Sales Performance Analysis (15 marks)

Calculate core KPIs: Total Sales, Average Daily Sales, Number of Stores, Number of Products, Number of Cities, Stockout Rate and Average Discount.

Identify Top 10 products by sales and compare performance across stores/cities.

Compare weekday vs weekend, holiday vs non-holiday, and activity/promotion vs normal periods.

For at least THREE findings, write both an Observation and an Insight.

Part D — Stockout and Potential Lost Opportunity (15 marks)

Calculate stockout frequency/rate and average stockout duration.

Identify products and stores with frequent stockouts.

Compare stockout days with non-stockout days using sales and at least two contextual variables such as discount, holiday or activity status.

Build a 2×2 product matrix using Sales (High/Low) and Stockout (High/Low).

Identify at least THREE High-Sales + High-Stockout products and explain why they may deserve management attention.

Part E — Promotion and External Factors (10 marks)

Create meaningful discount groups (for example: No, Low, Medium, High discount) and justify the thresholds.

Compare Average Sales, Stockout Rate and Number of Observations across discount groups.

Investigate whether stronger discounts are associated with higher stockout risk.

Choose ONE external factor (e.g., rainfall, temperature, humidity, holiday or activity) and analyse whether the pattern is consistent across relevant segments.

Explain why association does not automatically prove causation.

Part F — Interactive Business Dashboard (15 marks)

Create an interactive dashboard using Power BI, Tableau or Python. The dashboard should be equivalent to a maximum of THREE views/pages/screens.

View	Required focus	Minimum content
1. Executive Overview	Overall performance	Sales KPIs, stockout KPI, sales trend, product/store/city comparison
2. Inventory Risk	Availability and lost-opportunity risk	Top stockout products/stores, Sales vs Stockout, priority products
3. Decision Drivers	Promotion and external factors	Discount/promotion analysis plus at least one holiday/weather/activity analysis

Higher marks require useful interaction such as filtering, drill-down, dynamic comparison, informative tooltips, or decision-focused annotations. A visually attractive dashboard without business purpose will not receive high marks.

Part G — Decision Track (10 marks)

Each student must choose ONE track. This is a major source of differentiation between submissions.

Track	Decision question
A — Inventory Optimisation	Which products and stores should receive additional inventory?
B — Promotion Effectiveness	Where are discounts effective and where may they waste margin or create stockout risk?
C — Store Allocation	Should inventory be redistributed between stores or cities?
D — External Demand Drivers	How should weather, holidays or activities influence operational planning?

Part H — Unique Analytical Question (5 marks)

Define and answer ONE analytical question that is not directly answered by the compulsory tasks. It must be relevant to your selected Decision Track.

Do promoted products stock out faster?

Are weekends equally important in every city?

Does discount effectiveness vary by store or product group?

Which products show the strongest evidence of hidden demand?

Does rainfall affect different product groups differently ?

Part I — Management Recommendation and Limitations (10 marks)

Management can fund only ONE major initiative next month. Your final recommendation must include:

A clearly stated business question.

At least THREE relevant KPIs.

At least TWO visual pieces of evidence.

Observation: What happened ?

Insight: Why does it matter ?

Decision: What should management do ?

One important limitation, uncertainty or risk.

Required logic: Question → Evidence → Observation → Insight → Decision → Limitation

6. Optional Distinction Challenge (+ up to 10 bonus marks)

Students aiming for deeper analysis may complete ONE challenge. A complex model is not automatically better; the analysis must improve a business decision.

Challenge	Expectation
1. Hidden Demand	Estimate potential lost demand/sales during stockout periods using a clearly explained comparison or benchmark.
2. Promotion × Stockout	Investigate whether promotion/discount is associated with higher stockout risk and discuss confounding factors.
3. Product Segmentation	Create a decision-oriented segmentation using sales, stockout, discount and/or volatility.
4. Simple Predictive Extension	Use a simple model only if it supports a clear managerial decision; explain assumptions and limitations.

7. How the Assignment Differentiates Performance Levels

Indicative level	Typical evidence quality
Basic / Pass	Correct KPIs and descriptive charts; limited interpretation. Example: "Product A has the highest sales."
Competent	Useful comparisons and relevant business interpretation. Example: "Product A has high sales and high stockout frequency."
Strong	Multiple evidence sources, trade-offs and segment comparisons. Example: "High stockout may constrain observed sales, so more discount may not be appropriate."
Excellent	Critical reasoning, uncertainty and limits. Example: "The evidence suggests availability constrains observed sales, but lost-demand estimates depend on comparable-period assumptions and do not prove causal demand."

8. Deliverables

Deliverable	Requirement

Analysis file	SQL / KNIME / Excel / Python used for preparation and analysis. Must be understandable and reproducible.
Interactive dashboard	Power BI (.pbix), Tableau workbook/package, or Python dashboard/application files.
Management report	PDF, maximum 6 A4 pages excluding cover page.
Data preparation notes	PDF or Markdown, maximum 1 page, documenting cleaning decisions, derived variables and assumptions.
Individual video presentation	One 5–7 minute video (MP4 or an accessible LMS/approved link) explaining the analytical reasoning and decision-making process, challenges encountered, solutions adopted, key findings, final recommendation, limitations, and 2–3 takeaway points. The student must demonstrate part of the actual work rather than simply read the report.

8A. Individual Video Presentation

Each student must submit ONE individual video presentation of approximately 5–7 minutes. The purpose is to demonstrate that you understand and can defend your own analysis.

The video should cover the following:

1. Business problem — explain the selected business question and why it matters.
2. Analytical approach — summarise how you moved from raw data to the final recommendation, including data preparation, KPI selection, analysis performed, and dashboard design.
3. Challenges encountered — describe at least TWO meaningful challenges, such as data quality, dataset size, stockout interpretation, conflicting evidence, dashboard design, or tool limitations.
4. Solutions adopted — explain how you addressed those challenges and why those solutions were chosen.
5. Key findings and recommendation — present the strongest evidence and the ONE business action you recommend.
6. Limitations — state at least ONE important uncertainty, assumption, or limitation.
7. Takeaway points — conclude with 2–3 lessons learned from completing the assignment.

Important: The video should explain your analytical reasoning and decision-making process; it should not merely read the report. Show and explain part of your actual dashboard, SQL/KNIME/Python/Excel work, data-cleaning logic, KPI calculation, or analytical comparison.

9. Management Report Structure

1. Business Problem and chosen Decision Track
2. Dataset Understanding and Data Quality

3. Core Sales and Stockout Findings
4. Promotion / External-Factor Analysis
5. Unique Analytical Question
6. Dashboard Evidence
7. Final Recommendation
8. Limitations and Conclusion

The report should focus on evidence and reasoning, not on screenshots of every query or every workflow node.

10. File Naming and Submission Instructions

Create one folder using the following naming convention:

```
StudentID_Midterm
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Recommended individual file names:

StudentID_Report.pdf
StudentID_DataPreparation.pdf (or .md)
StudentID_Analysis.sql / .knwf / .xlsx / .ipynb / .py
StudentID_Dashboard.pbix OR StudentID_Dashboard.twbx OR StudentID_Dashboard/ (Python dashboard files)

Include the individual video as StudentID_Video.mp4, or include a text/PDF file named StudentID_VideoLink containing the approved accessible video URL. Then compress the entire folder into ONE ZIP file and submit it through the LMS:

```
StudentID_Midterm.zip
```

Check that the dashboard opens correctly.
Check that all analysis files run or can be inspected without broken references.
If external files are required, include them or provide clear relative paths.
Do not submit only screenshots of the dashboard or analysis. The video must be accessible to the lecturer at the time of marking.

11. Deadline and Late Submission

Deadline: 23:59, Sunday, 11 October 2026.

Late period	Suggested penalty
Up to 24 hours	-10%
24–48 hours	-20%
More than 48 hours	Not accepted unless prior approval applies under course/university regulations.

12. Assessment Rubric

The midterm is graded by submitted assessment component. The weights below total 100% and apply equally regardless of whether the dashboard is built in Power BI, Tableau, or Python.

Academic-integrity concerns may also be handled under course and university regulations. The 5% criterion below evaluates transparent attribution, ownership, and the ability to explain/reproduce submitted work; it does not limit any separate penalties that may apply to confirmed misconduct.

Assessment Component	Weight	Criteria for Full-Credit Performance
1. Analysis	35%	Frames a relevant business question; correctly understands the FreshRetailNet-50K data; applies appropriate data-quality checks and transformations; calculates accurate KPIs; completes the required sales, stockout, promotion/external-factor analyses; answers the selected Decision Track and Unique Analytical Question; uses evidence rather than unsupported claims; and documents assumptions that affect interpretation.
2. Interactive Dashboard	20%	Uses Power BI, Tableau, or a Python-based dashboard effectively. The dashboard is decision-focused, accurate, readable, and interactive; includes the required executive, inventory-risk, and decision-driver views; uses appropriate filters/drill-downs or equivalent interaction; and highlights evidence that supports the final recommendation rather than merely displaying many charts.
3. Management Report	20%	Presents a concise and professional report (maximum 6 A4 pages excluding cover) with a clear business problem, selected Decision Track, key evidence, Observation → Insight → Decision reasoning, final recommendation, and limitations. Visual evidence is interpreted correctly and the recommendation is specific, actionable, and supported by multiple relevant KPIs.
4. Data-Preparation Notes	10%	Provides a clear, reproducible record (maximum 1 page) of filtering/sampling, missing-value treatment, duplicate handling, unusual-value decisions, derived variables, assumptions, and any transformations that materially affect the analysis. Decisions are justified in business/analytical terms rather than listed without explanation.
5. Individual Video Presentation	10%	Submits a 5–7 minute individual presentation that explains the business problem, analytical approach, at least two meaningful challenges, solutions adopted, key findings, recommendation, limitations, and 2–3 takeaway points. The student demonstrates part of the actual dashboard/analysis and shows understanding rather than simply reading the written report.
6. Academic Integrity & Ownership	5%	Work is independently produced and properly attributed where external sources, code, AI tools, templates, or references are used. The student can explain and reproduce submitted calculations, transformations, dashboard logic, and recommendations. File contents are internally consistent and do not contain unexplained copied material or fabricated evidence.
TOTAL	**100%**	

13. Short Oral Defense

The lecturer may conduct a short individual oral defense when needed. The student may be asked to explain a KPI, cleaning decision, visual, analytical method, dashboard interaction, or recommendation.

- Why did you choose this KPI?
- Why did you classify this product as high risk?
- What would change your recommendation?
- What does your dashboard fail to show?
- Can your analysis prove causation?
- Why should management trust this recommendation?

14. Academic Integrity and Use of AI Tools

Students may use course materials, official software documentation, dataset documentation and AI tools for learning support. However, each student is individually responsible for understanding and defending every query, transformation, KPI, dashboard calculation and recommendation submitted.

If a student cannot explain or reproduce a submitted element, the corresponding work may receive limited or no credit.

15. Final Question

If management can take only ONE action next month, what should it do, what evidence supports that decision, and what uncertainty should management still be aware of?

The goal of this midterm is not to produce the largest number of charts. The goal is to make a business decision that can be defended with relevant, trustworthy evidence.